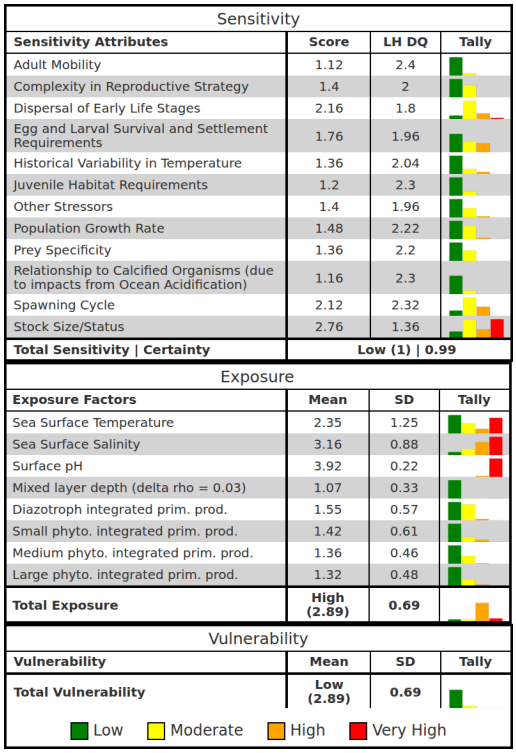
Chub mackerel (*Scomber colias*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** High
- **Vulnerability:** Low



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.12	2.4	
Complexity in Reproductive Strategy	1.4	2	
Dispersal of Early Life Stages	2.16	1.8	
Egg and Larval Survival and Settlement Requirements	1.76	1.96	
Historical Variability in Temperature	1.36	2.04	
Juvenile Habitat Requirements	1.2	2.3	
Other Stressors	1.4	1.96	
Population Growth Rate	1.48	2.22	
Prey Specificity	1.36	2.2	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.16	2.3	
Spawning Cycle	2.12	2.32	
Stock Size/Status	2.76	1.36	
Total Sensitivity | Certainty	**Low (1) | 0.99**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Temperature	2.35	1.25	
Sea Surface Salinity	3.16	0.88	
Surface pH	3.92	0.22	
Mixed layer depth (delta rho = 0.03)	1.07	0.33	
Diazotroph integrated prim. prod.	1.55	0.57	
Small phyto. integrated prim. prod.	1.42	0.61	
Medium phyto. integrated prim. prod.	1.36	0.46	
Large phyto. integrated prim. prod.	1.32	0.48	
Total Exposure	**High (2.89)**	**0.69**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2.89)**	**0.69**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Chub mackerel. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Not Available	Low	-
Exposure	Not Available	High	-
Vulnerability	Not Available	Low	-
Directional Effect	Not Available	Neutral	-

Stocks

Chub mackerel is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

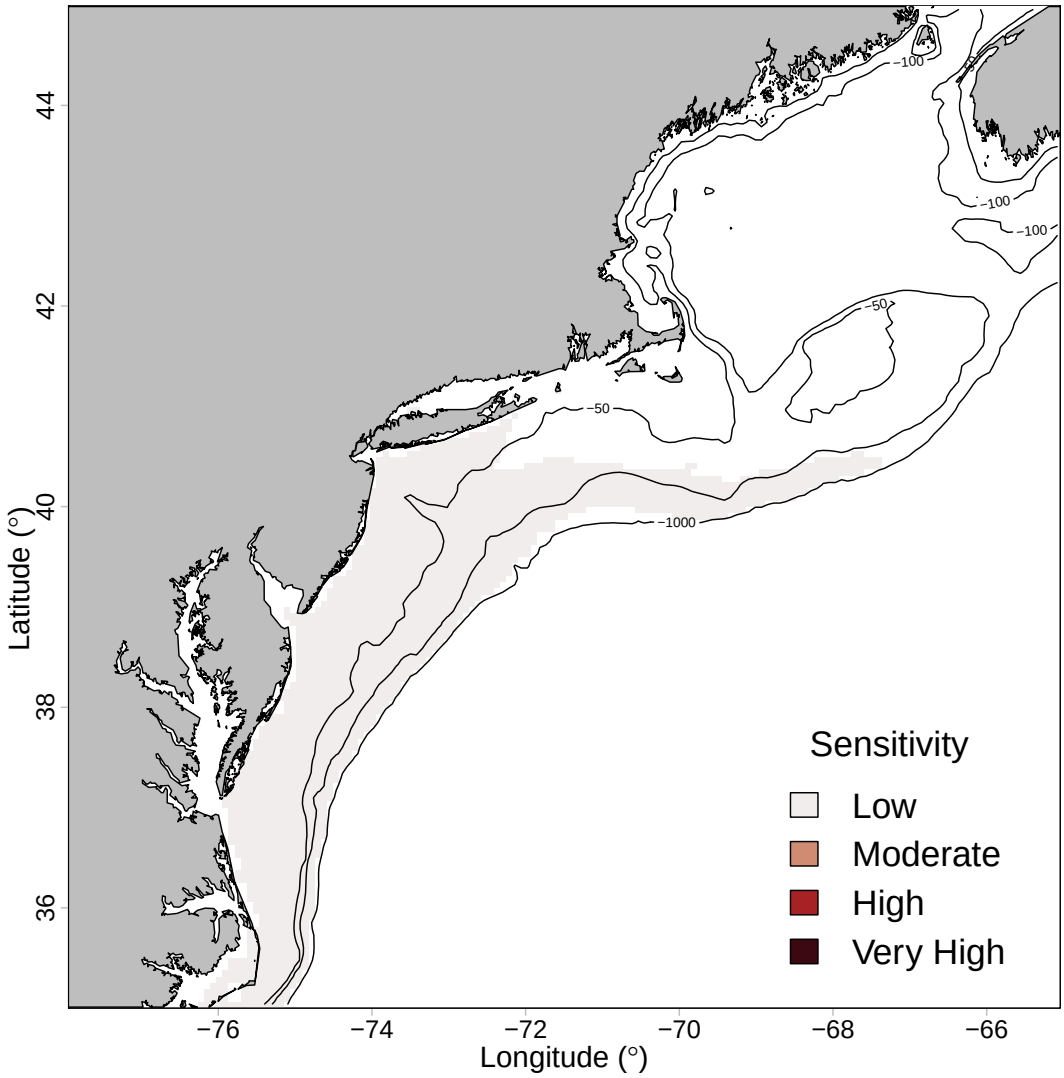


Figure 2: Chub mackerel Sensitivity on the Northeast Shelf

Chub mackerel were given a **Low** sensitivity score, with 99% certainty according to bootstrap analysis (Figure 1). The only attribute that received a score greater than 2.5 was Stock Size. This is likely because no stock assessment for chub mackerel exists (Mid-Atlantic Fishery Management Council (MAFMC) 2025).

Chub mackerel were not assessed in the 2016 Climate Vulnerability Assessment, nor any other recent Assessment (Hare et al. 2016).

Exposure:

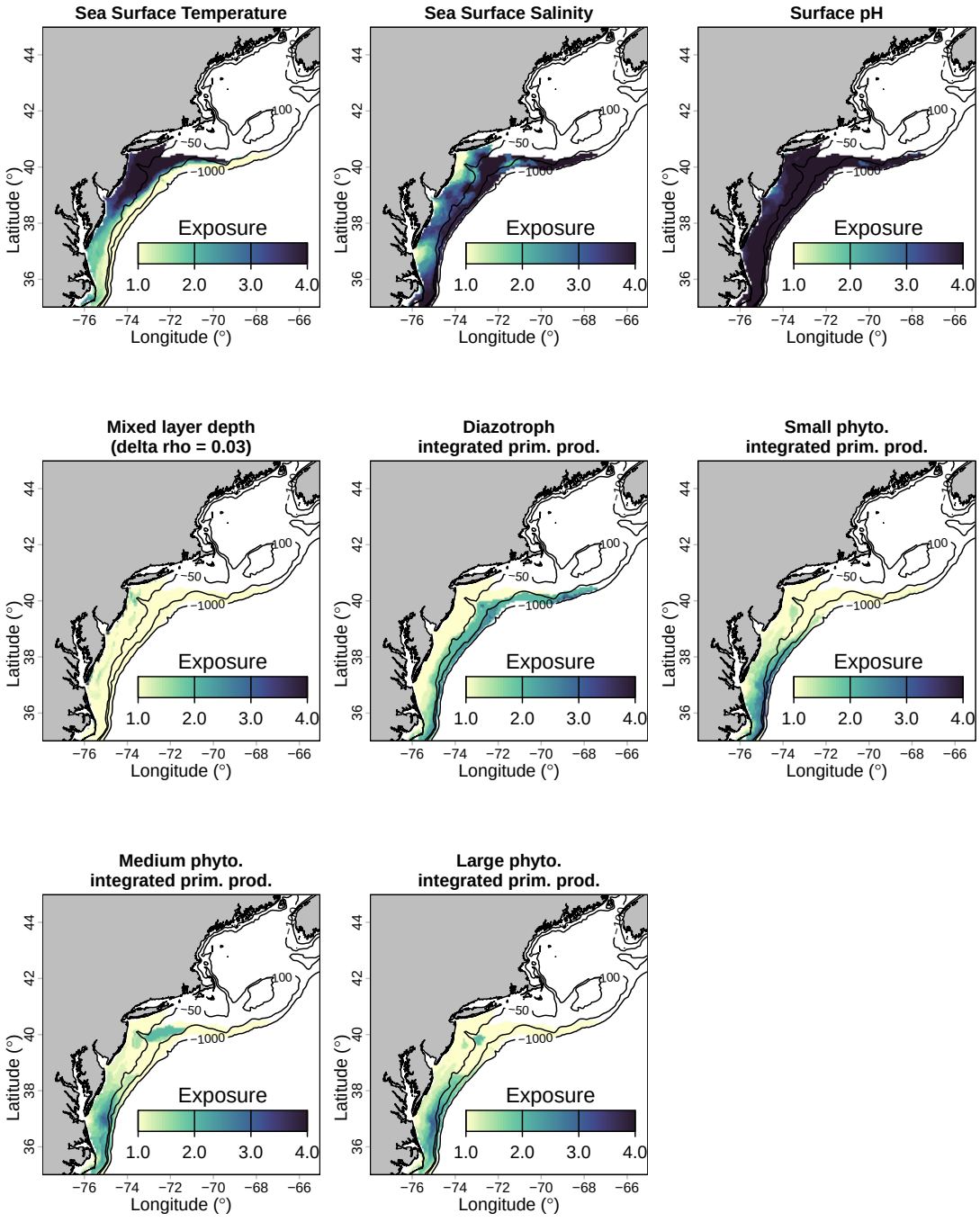


Figure 3: Maps of each Exposure Factor for Chub mackerel. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for Chub mackerel was **High** (2.89 +/- 0.69) (Figure 1). Exposure was High throughout most of the species range, but was Very High south of Long Island and Cape Cod, and Low off the coast

of Chesapeake Bay (Figure 4). This was driven by surface temperature, salinity, and pH exposures, plus high exposures in phytoplankton productivity variables (all size classes) around Hudson Canyon and off the coasts of Delaware and Virginia (Figure 3).

Chub mackerel were not assessed in the 2016 Climate Vulnerability Assessment, nor any other recent Assessment (Hare et al. 2016).

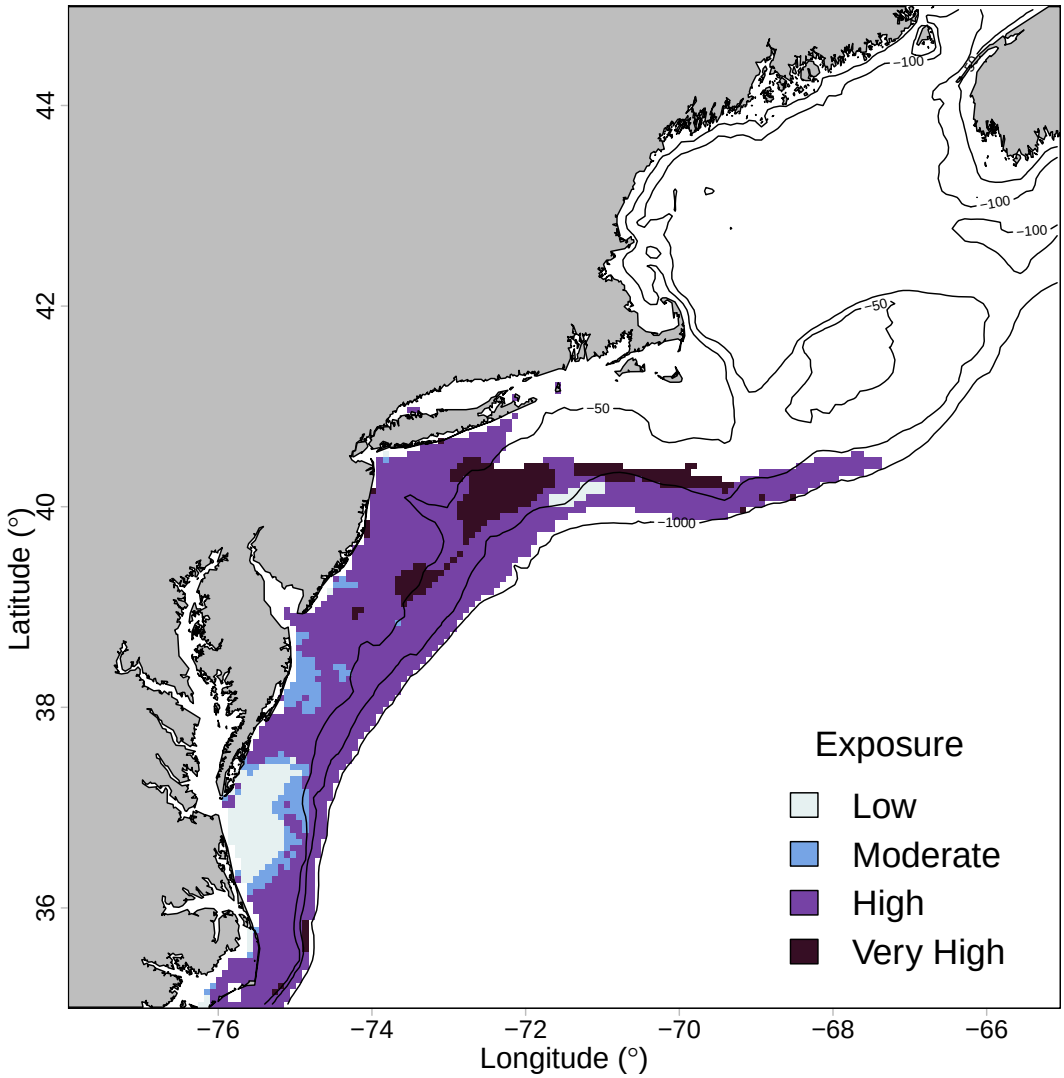


Figure 4: Total Exposure on the Northeast Shelf for Chub mackerel. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

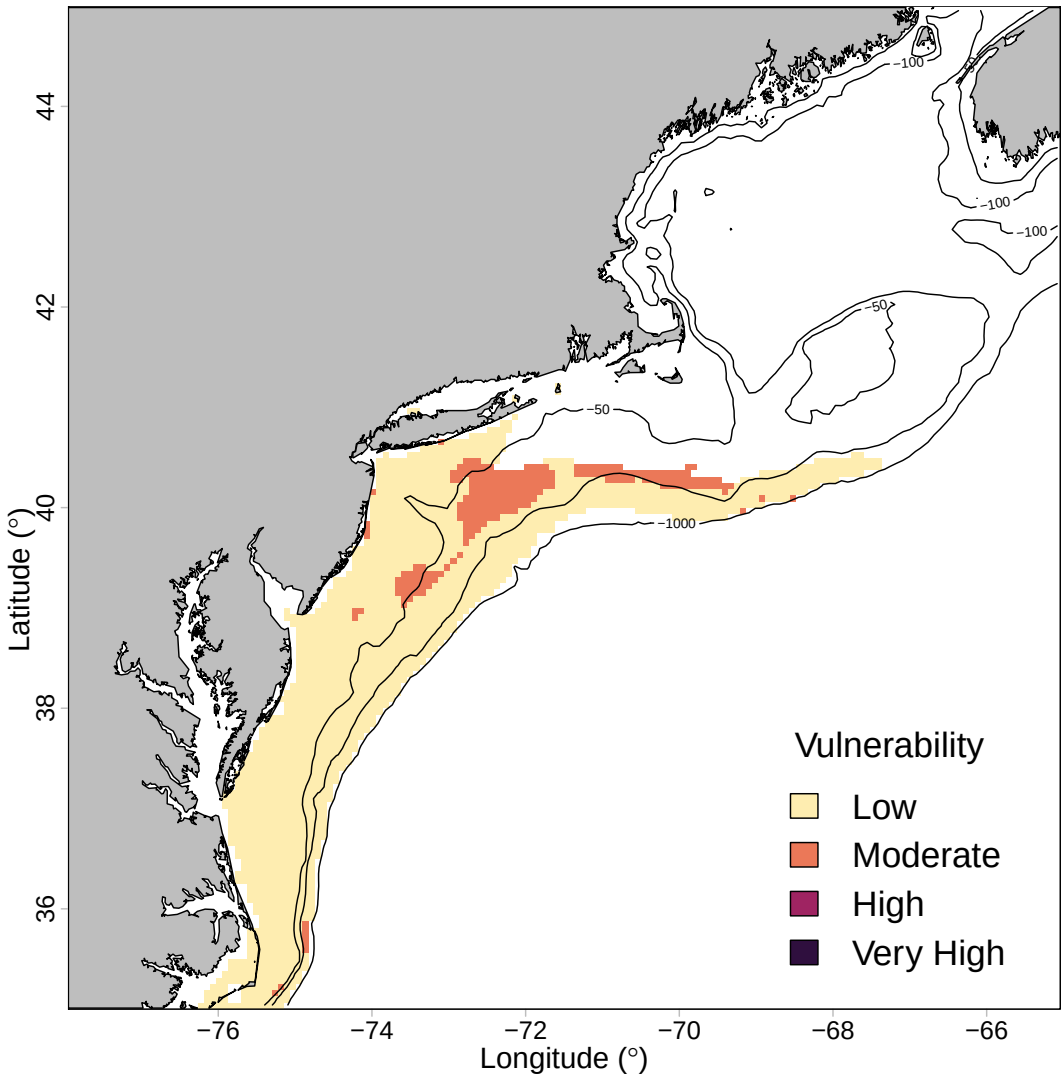


Figure 5: Vulnerability on the Northeast Shelf for Chub mackerel. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Chub mackerel was **Low** (2.89 +/- 0.69) due to Low sensitivity scores (Figure 1). Vulnerability was Low throughout the species' range, except south of Long Island and Cape Cod where Vulnerability was Moderate (Figure 5).

Chub mackerel were not assessed in the 2016 Climate Vulnerability Assessment, nor any other recent Assessment (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

Only 68% of data quality scores were greater than or equal to 2, indicating poor data quality. Data gaps (mean attribute data quality score < 2) included Complexity in Reproductive Strategy, Dispersal of Early Life Stages, Egg and Larval Survival and Settlement Requirements, Other Stressors, and Stock Size/Status (Figure 1).

Model Confidence Score:

The chub mackerel ensemble model received a mean model confidence score of 2.1 (+/- 0.6), indicating moderate confidence in the final model. Experts thought that the model generally described chub mackerel distributions well, but the low sample size drove scores down. The ensemble AUC score for chub mackerel was 0.89 indicating that the final model had good predictive performance.

Additional Metrics

Distribution Shifts:

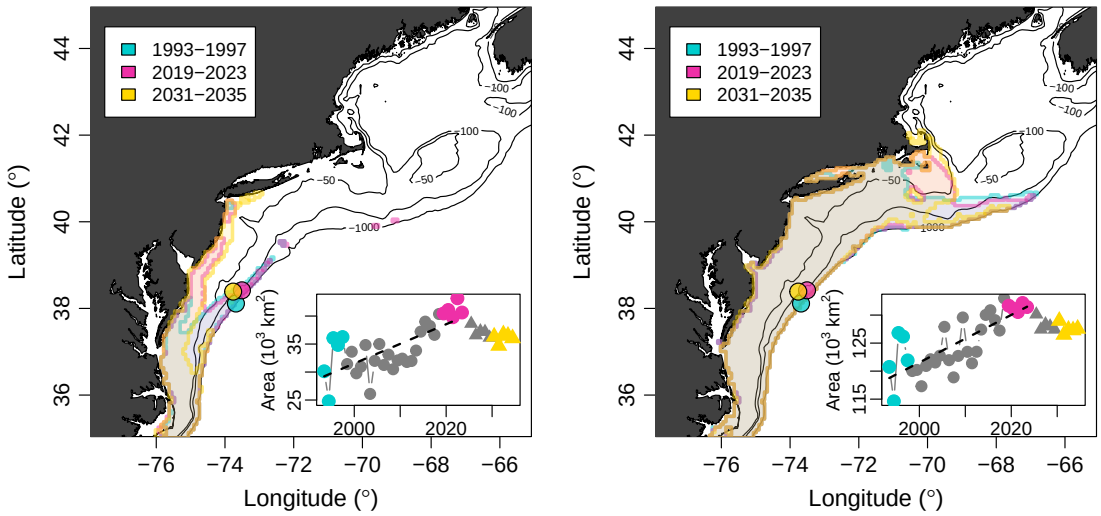


Figure 6: Distribution shifts on the Northeast Shelf for Chub mackerel. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Chub mackerel experienced significant habitat expansion between 1993 and 2023. Core habitats expanded further north along the coast of New Jersey and their range expanded further east along the coast of

southern New England (Figure 6). The weighted centroid shifted 27 km north-northeast between 1993 and 2023.

Under near future climate conditions, chub mackerel habitat is expected to remain similar. Chub mackerel are predicted to lose habitat on the continental shelf break, but gain habitat nearshore and on the shelf itself as the northern and eastern edges of their distribution continue to shift further north and east respectively (Figure 6). The weighted centroid is predicted to shift 34 km westward, reflecting the increase in inshore habitat use.

Chub mackerel were not assessed previously so no previous ranks are available.

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	33,926	124,707	5.61	6.49
2019-2023	37.2 km NNE	41,283 (+7,356)	131,768 (+7,060)	5.80 (+0.19)	6.41 (-0.08)
2031-2035	33.5 km W	36,136 (-5,146)	127,610 (-4,157)	5.86 (+0.06)	5.93 (-0.48)

Directional Effect:

Experts predicted that climate change would have a **neutral** effect on chub mackerel. Expert certainty was 87%. Variations in abundance have been observed and are likely due to oceanographic processes due to their pelagic lifestyle, but the exact drivers are unclear (Hern'andez and Ortega 2000).

Climate Effects on Abundance and Distribution

Fluctuations in Atlantic chub mackerel abundance have been reported in the Mid-Atlantic and New England, and may result from environmental influences such as temperature and upwelling. Oceanographic processes likely influence recruitment and distribution, as all life stages of chub mackerel are fully pelagic (Hern'andez and Ortega 2000). In the eastern Atlantic, chub mackerel distributions have shifted northward, with a delayed and prolonged spawning season with increasing latitude (Dom'inguez-Petit et al. 2022). In the Alboran Sea, chub mackerel abundance is vulnerable to variations in sea surface temperature, salinity, currents, and wind patterns (Ayman et al. 2024).

Life History Synopsis

Atlantic chub mackerel is a schooling pelagic species ranging from Nova Scotia to Argentina, including the Gulf of America, and are found on the continental shelf at depths of 250-300m (Mid-Atlantic Fishery Management Council (MAFMC) 2025). They are batch spawners, spawning January through April from North Carolina through the Gulf of America (Daley 2018). Spawning generally occurs in waters greater than 10°C and is possibly triggered by water temperatures between 15- 20°C (Hern'andez and Ortega 2000). Chub mackerel eggs develop for approximately 53-56 hrs at 18-20°C. Larval stages last between 16-24 days, developing faster at higher temperatures (Hern'andez and Ortega 2000). Atlantic

chub mackerel grow rapidly during the first year of life, reaching maturity around 2 years of age (Daley 2018) and can live up to 13 years (Hern'andez and Ortega 2000). Chub mackerel are generalist predators of zooplankton and other pelagic prey including mysids, euphausids, decapod larvae, copepods, amphipods, small fish, cephalopods. Cannibalism has been observed in this species (Hern'andez and Ortega 2000). There is little data on chub mackerel predation in the Northwest Atlantic. The Mid-Atlantic Fishery Management Council funded a study on diet composition of marlins and tunas and found very few chub mackerel in the stomachs of yellowfin tuna and white marlin (Mid-Atlantic Fishery Management Council (MAFMC) 2025). In 2019, the Mid-Atlantic Fishery Management Council approved Amendment 21 to the Mackerel, Squid, and Butterfish Fishery Management Plan. This Amendment established conservation and management measures for Atlantic chub mackerel in the Northwest Atlantic (Mid-Atlantic Fishery Management Council (MAFMC) 2019). There is currently no stock assessment of chub mackerel in the Mid-Atlantic. Biomass is assumed to be at a sustainable level based on stock productivity and low fishery capacity (Mid-Atlantic Fishery Management Council (MAFMC) 2025).

Supplemental Scores

Exposure Timeseries

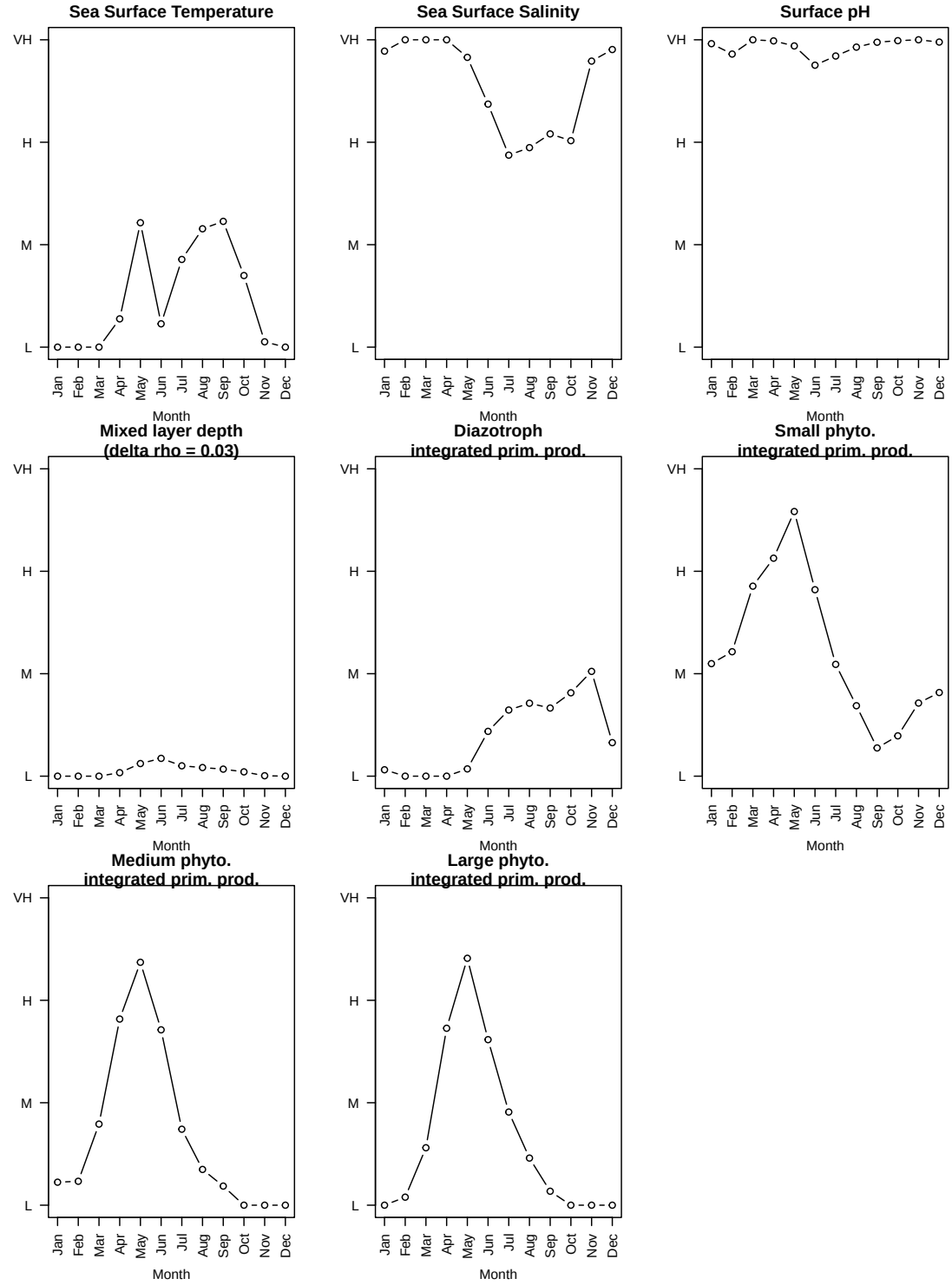


Figure 7: Monthly Timeseries of each Exposure Factor for Chub mackerel. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure ranged from Moderate to Very High (Figure 8). Scores were Very High in May due to coincident spikes in phytoplankton biomass exposure across size classes. Exposure declined to Moderate in July and August due to declines in surface salinity exposure (Figure 7).

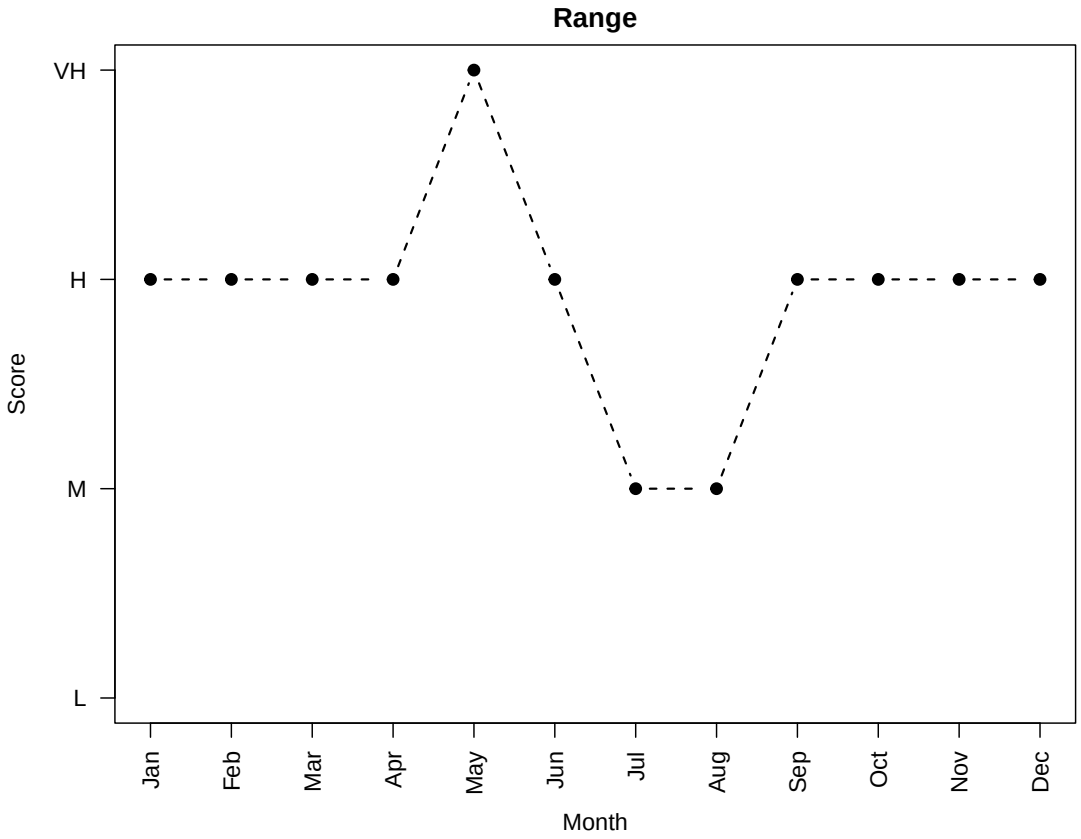


Figure 8: Timeseries of Total Exposure for Chub mackerel

Exposure due to Significant Factors

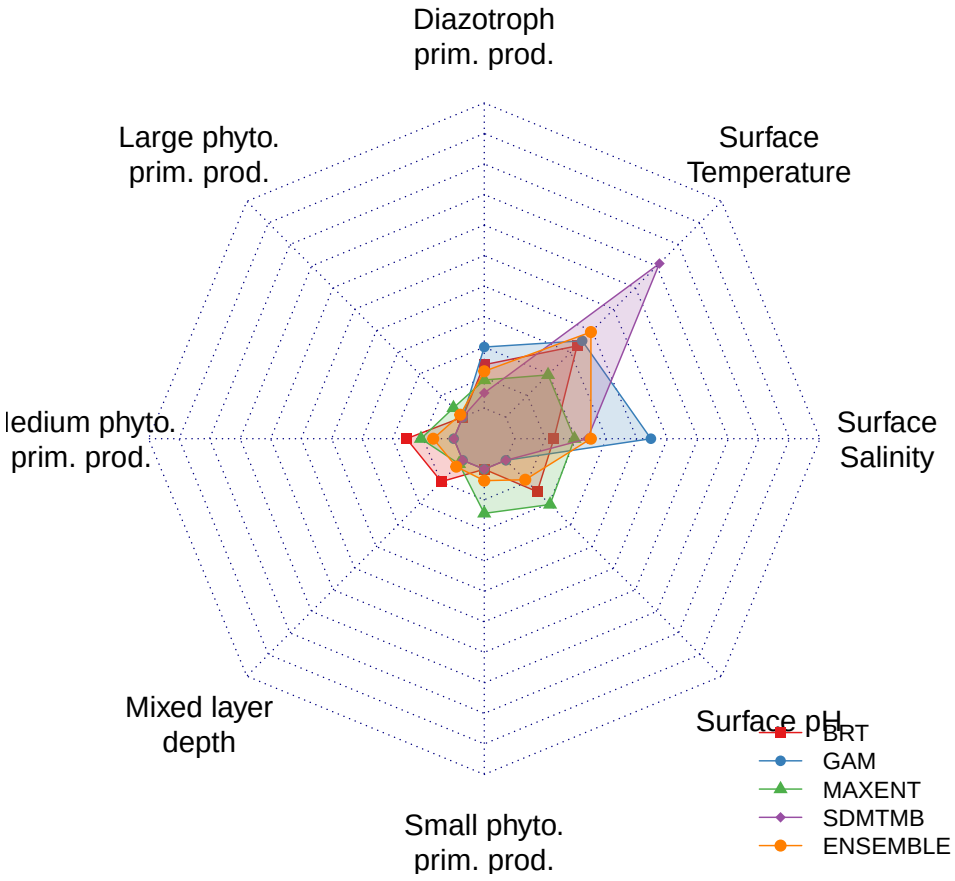


Figure 9: Dynamic Factors driving Chub mackerel distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Sea surface temperature, surface salinity, and diazotroph primary productivity were important dynamic drivers of chub mackerel distributions on the Northeast Shelf (Figure 9). Chub mackerel are generalist zooplankton predators and oceanographic processes play a role in spawning and their distribution (Hern'andez and Ortega 2000). When only these factors are included in the exposure calculation, mean exposure drops from High to Moderate (1.54 +/- 0.83) (Figure 11). Exposure in this scenario is High off the coasts of Long Island and New Jersey, where high surface temperature and salinity exposures overlap, but Low elsewhere (Figure 10).

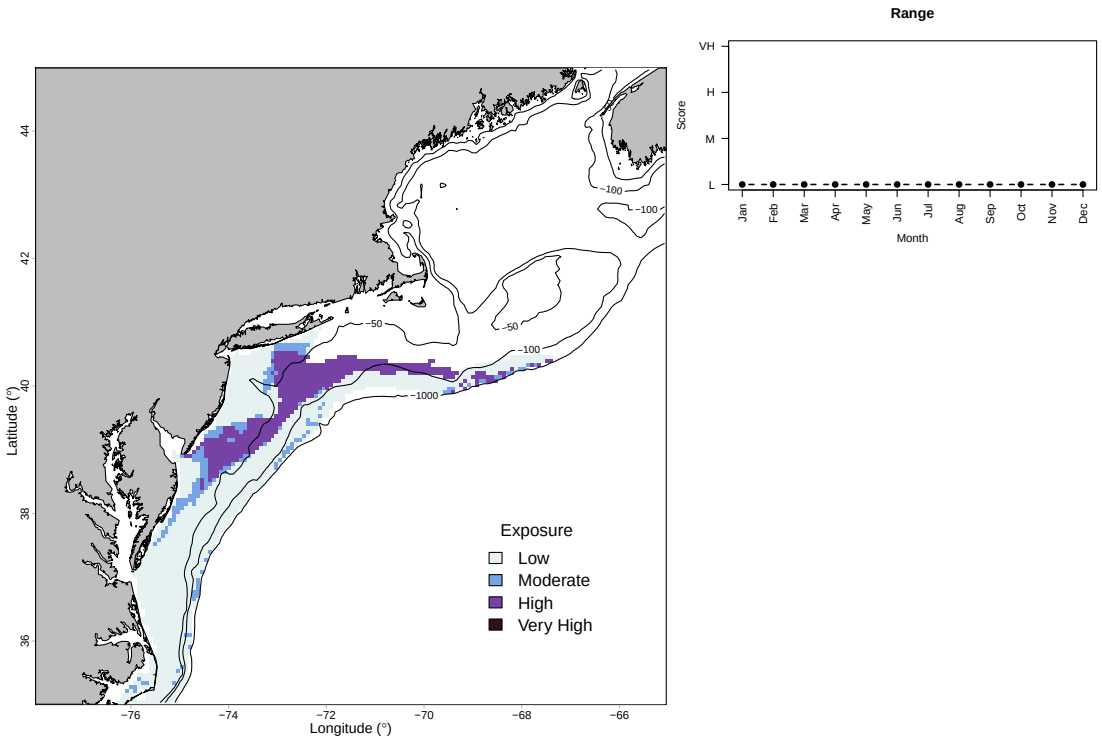
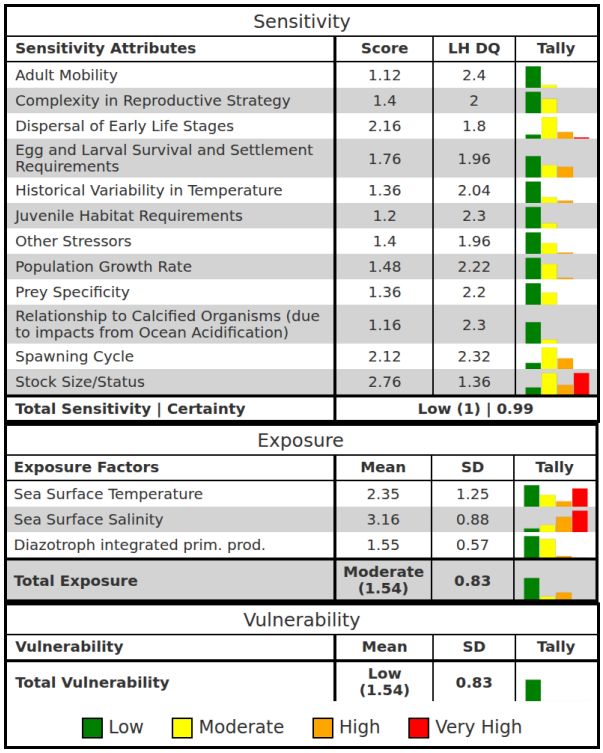


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Chub mackerel distributions



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.12	2.4	
Complexity in Reproductive Strategy	1.4	2	
Dispersal of Early Life Stages	2.16	1.8	
Egg and Larval Survival and Settlement Requirements	1.76	1.96	
Historical Variability in Temperature	1.36	2.04	
Juvenile Habitat Requirements	1.2	2.3	
Other Stressors	1.4	1.96	
Population Growth Rate	1.48	2.22	
Prey Specificity	1.36	2.2	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.16	2.3	
Spawning Cycle	2.12	2.32	
Stock Size/Status	2.76	1.36	
Total Sensitivity | Certainty	**Low (1) | 0.99**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Temperature	2.35	1.25	
Sea Surface Salinity	3.16	0.88	
Diazotroph integrated prim. prod.	1.55	0.57	
Total Exposure	**Moderate (1.54)**	**0.83**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (1.54)**	**0.83**	

Figure 11: As in Figure 1 for Chub mackerel, but with only significant Exposure factors and the resulting Vulnerability Scores.

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